

A Peek Beyond the Point

CHAPTER

03

Learning Outcomes...

- The Need for Smaller Units
- A Tenth Part
- A Hundredth Part
- Decimal Place Value
- Use of Decimals (Units of Measurement)
- Locating and Comparing Decimals
- Addition and Subtraction of Decimals
- Decimals in Daily Life and History

Introduction

In our daily life, we often come across measurements that are too small like the length of a screw, the size of a pencil, the height of a plant, or even the amount of petrol we buy cannot always be written as exact whole units. That is why we divide each unit into smaller equal parts called decimals. Decimals help us measure and compare these tiny differences more accurately. We use them while buying vegetables, measuring cloth, calculating money, or measuring our body parts. Understanding decimals and learning how to add and subtract them makes our daily calculations easier and more precise.

The Need for Smaller Units

The need for smaller units arises when the objects we measure are very small or when their sizes differ only slightly. Whole units (like 1 cm or 1 meter) are not enough to show such small differences. By dividing a unit into smaller equal parts—like millimeters, tenths, or hundredths—we can measure objects more accurately and compare their lengths correctly.

A Tenth Part

Reading and Writing Tenths

To understand and measure small differences accurately, we divide 1 centimeter into 10 equal parts. Each part is called one-tenth of a centimeter ($\frac{1}{10}$ cm). So, 2.7 cm means 2 full centimeters + 7 one-tenths of a centimeter and 3.2 cm means 3 full centimeters + 2 one-tenths of a centimeter. These smaller parts help us to measure more accurately, especially when values are close to each other.

Let's try to measure the length of a crayon.

On a ruler, the space between two whole number marks (such as from 4 to 5 cm) is divided into 10 equal parts. Each part represents $\frac{1}{10}$ of a centimeter. To read a measurement like $4\frac{6}{10}$ cm, we move



from 0 to 4 cm, and then count six small divisions (tenths) beyond that. So, the total length of crayon is 4 full centimeters and 6 tenths, written as $4\frac{6}{10}$ cm and read as four and six-tenth centimeters.

Two Ways of Expressing Lengths

Suppose a thin metallic rod has a length of $2\frac{5}{10}$ units. This can be understood as 2 complete units

and 5 additional one-tenth units. That is : $(2 \times 1) + \left(5 \times \frac{1}{10}\right) = 2 + \frac{5}{10} = 2\frac{5}{10}$ units

We can also express this length using tenths alone i.e., the length of sheet is 25 one tenths units, since every 10 tenths combine to form one whole unit.

$$25 \times \frac{1}{10} = \frac{25}{10} = \frac{10}{10} + \frac{10}{10} + \frac{5}{10} = 1 + 1 + \frac{5}{10} = 2 \text{ and } 5 \text{ one-tenths}$$

These two ways of representing lengths mixed numbers $\left(2\frac{5}{10}\right)$ and fractions $\left(\frac{25}{10}\right)$ both are equally valid and useful in understanding and comparing measurements.

Addition and Subtraction of Tenths and Whole Numbers

When working with measurements that involve whole numbers and tenths, we can either add them part by part or convert everything into tenths.

Part-by-Part Method

Suppose we need to add $3\frac{5}{10}$ and $2\frac{7}{10}$.

Add whole numbers : $3 + 2 = 5$

Add tenths : $\frac{5}{10} + \frac{7}{10} = \frac{12}{10}$

Now, convert $\frac{12}{10}$ into a whole and a fraction: $\frac{12}{10} = 1 + \frac{2}{10}$. So, required sum = $5 + 1 + \frac{2}{10} = 6\frac{2}{10}$

Convert to Tenth Method

$$3\frac{5}{10} = \frac{35}{10} \text{ and } 2\frac{7}{10} = \frac{27}{10}. \text{ So, } \frac{35}{10} + \frac{27}{10} = \frac{62}{10} = 6\frac{2}{10}$$

Similarly, we can subtract the two numbers but sometimes, when subtracting tenths, the number of tenths in the top number is smaller than the number of tenths in the bottom number. In such cases we need to borrow from the whole number.

Subtraction with Borrowing

Suppose we need to subtract $3\frac{7}{10}$ from $6\frac{5}{10}$. Break into parts: $\left(6 + \frac{5}{10}\right) - \left(3 + \frac{7}{10}\right) = (6 - 3) + \left(\frac{5}{10} - \frac{7}{10}\right)$

Here, $\frac{5}{10}$ is less than $\frac{7}{10}$, so we borrow 1 whole (which is $\frac{10}{10}$) from 6. So, 6 becomes 5 and $\frac{5}{10}$ becomes $\frac{15}{10}$.

$$\text{So, } \left(5 + \frac{15}{10}\right) - \left(3 + \frac{7}{10}\right) = (5 - 3) + \left(\frac{15}{10} - \frac{7}{10}\right) = 2\frac{8}{10}$$

ILLUSTRATIONS

1 Two colored ribbons are being measured for decoration.

Ribbon A: $4\frac{3}{10}$ meters

Ribbon B: $4\frac{7}{10}$ meters, which is longer?

Soln.: We compare the whole numbers in length of ribbon A and B; both are 4.

Now compare the tenths: Since, $\frac{3}{10} < \frac{7}{10}$

So, Ribbon B is longer.

2 Riya is sewing two pieces of fabric together.

First piece has length $2\frac{6}{10}$ meters and second piece has length $3\frac{5}{10}$ meters. What is the total length of the fabric sewed together?

Soln.: Total length of fabric = $2\frac{6}{10} + 3\frac{5}{10}$

$$= (2 + 3) + \left(\frac{6}{10} + \frac{5}{10}\right)$$

[Break it into whole numbers and tenths]

$$= 5 + \frac{11}{10} = 5 + \frac{10}{10} + \frac{1}{10} = 5 + 1 + \frac{1}{10} = 6\frac{1}{10}$$

3 Arjun had a wooden plank $6\frac{1}{10}$ meters long. He cut off $3\frac{7}{10}$ meters for a shelf. How much is left?

Soln.: We need to subtract $3\frac{7}{10}$ from $6\frac{1}{10}$.

Tenths: $\frac{1}{10}$ is less than $\frac{7}{10}$ so, borrow 1 from 6.

So,

6 becomes 5 and $\frac{1}{10}$ becomes $\frac{11}{10}$

Length of remaining plank

$$= 5\frac{11}{10} - 3\frac{7}{10} = (5 - 3) + \left(\frac{11}{10} - \frac{7}{10}\right) = 2\frac{4}{10} \text{ metres}$$

A Hundredth Part

Sometimes, even dividing a unit into tenths is not enough to measure tiny differences. In such cases, we divide each one-tenth into 10 **smaller parts**. These parts are called hundredths because there are 100 parts of them in a full unit.

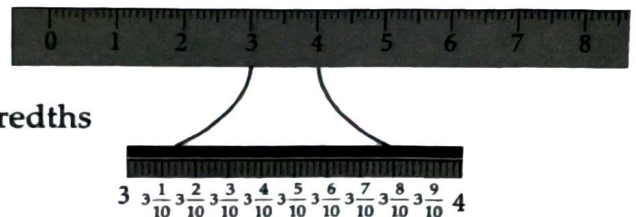
For example; Sara had a ribbon that was $6\frac{7}{10}$ meters long. She folded it exactly in half to make two equal pieces.

Now, what is the length of one piece?

Half of 6 is 3 and Half of $\frac{7}{10}$ is 3 tenths and 5 hundredths

So, the new length is $3\frac{3}{10}$ and $\frac{5}{100}$ metres.

We write this as $3\frac{3}{10}\frac{5}{100}$ read as 3 units and 3 one-tenths and 5 one-hundredths.



Understanding Hundredths

One-tenth can be divided into 10 equal parts

$$\text{So, } \frac{1}{10} = \frac{10}{100}$$

Therefore, 1 unit = 100 hundredths

These hundredths allow us to express even smaller measurements more clearly. There are different ways to express the hundredths.

For example : The length $1\frac{1}{10}\frac{4}{100}$ has 1 unit, 1 tenth $\left(\frac{1}{10}\right)$ and 4 hundredths $\left(\frac{4}{100}\right)$. This can be written as

$$1\frac{14}{100} \text{ (one and fourteen hundredths) or } \frac{114}{100} \text{ (as a fraction)}$$

All of them represent the same quantity.

Adding and Subtracting Mixed Numbers with Tenths and Hundredths

When adding numbers that include whole parts, tenths, and hundredths, we can break the parts and add them step by step.

- (i) Add the whole numbers separately and tenths together.
- (ii) Add the hundredths together.
- (iii) If the sum of hundredths is more than 10, convert it into tenths.
- (iv) If the tenths add up to 10 or more, convert them into whole units.

Note: 10 hundredths = 1 tenth and 10 tenths = 1 whole.

Similarly, we can do the subtraction. If the top number has fewer hundredths or tenths, we borrow from the next higher unit (borrow 1 tenth = 10 hundredths or 1 whole = 10 tenths). This allows us to subtract smoothly by converting a larger unit into smaller parts when needed.

ILLUSTRATIONS

- 4** Compare the lengths and identify the shortest and longest.

$$\frac{65}{100}, 5\frac{5}{10}, \frac{54}{100}, 4\frac{6}{10}\frac{4}{100}$$

Soln.: Let us convert the given numbers into hundredths

$$5\frac{5}{10} = \frac{55}{10} = \frac{550}{100} \text{ and } 4\frac{6}{10}\frac{4}{100} = 4\frac{64}{100} = \frac{464}{100}$$

$$\text{Now, } 54 < 65 < 464 < 550$$

So, $\frac{54}{100}$ is the shortest length and $5\frac{5}{10}$ is the longest length.

- 5** Meera walked $2\frac{7}{10}$ kilometers on Monday and $1\frac{45}{100}$ kilometers on Tuesday. How far did she walk in total?

Soln.: Firstly convert tenths to hundredths:

$$\frac{7}{10} = \frac{70}{100}. \text{ So, } 2\frac{7}{10} = 2\frac{70}{100}$$

Total distance walked by Meera

$$= 2\frac{70}{100} + 1\frac{45}{100} = 3\frac{115}{100} = 3 + \frac{115}{100} = 3 + 1\frac{15}{100} = 4\frac{15}{100}$$

- 6** Subtract $3\frac{5}{10}\frac{8}{100}$ from $6\frac{2}{10}\frac{3}{100}$.

Soln.: Let us convert the numbers to hundredths

$$6\frac{2}{10} = 6\frac{20}{100}. \text{ So, } 6\frac{20}{100}\frac{3}{100} = 6\frac{23}{100}$$

$$\text{Also, } 3\frac{5}{10} = 3\frac{50}{100}. \text{ So, } 3\frac{50}{100}\frac{8}{100} = 3\frac{58}{100}$$

$$\begin{aligned} \text{Now, } 6\frac{23}{100} - 3\frac{58}{100} &= \left(5 + 1 + \frac{23}{100}\right) - \left(3 + \frac{58}{100}\right) \\ &= \left(5 + \frac{123}{100}\right) - \left(3 + \frac{58}{100}\right) = 2 + \frac{65}{100} = 2\frac{65}{100} \end{aligned}$$

Decimal Place Value

Splitting Units for Accurate Measurements

To measure more precisely, we can break 1 whole unit into 10 equal parts. Each part is called one-tenth. If needed, we can go further. Each tenth can be split into 10 smaller parts, called hundredths. Each hundredth can be split into 10 even smaller parts, called thousandths and so on.

Importance of 10 in Place Value

The decimal system is based on 10. Each place value is 10 times the value of the place to its right, or $\frac{1}{10}$ of the value of the place to its left.

For example; 10 ones = 1 ten, 10 tens = 1 hundred, 10 hundreds = 1 thousand

Naming Decimal Fractions

Tenths: $\frac{1}{10}$ of a unit; Hundredths: $\frac{1}{100}$ of a unit ; and Thousandths: $\frac{1}{1000}$ of a unit

These are all fractions, showing parts of a whole. But writing and working with so many fractions can become difficult, especially in longer calculations. That's why we have a shorter way to write them using a special dot called a point, which helps us to show these small parts in our place value system. This point is called the decimal point.

To Read a Decimal Number

A decimal or a decimal number contains a whole number part and a decimal number part separated by a dot (·).

For example; In 356.47, 356 is the whole part and 47 is the decimal part. It is read as 'Three hundred fifty six point four seven'.

Decimal Fractions

The fractions in which the denominators are 10, 100, 1000, etc..., are known as decimal fractions.

Place Value of Decimal Numbers

In the whole number part, if we move from right to left by 1 place, the place value of a digit increases by 10 times. In the decimal part, if we move from left to right by 1 place, the value of a digit increases by $\frac{1}{10}$ times.

See the following place value chart for 4563.427, 879.34 and 6284.932

Thousands 1000	Hundreds 100	Tens 10	Ones 1	Decimal point	Tenths $\frac{1}{10}$	Hundredths $\frac{1}{100}$	Thousandths $\frac{1}{1000}$
4	5	6	3	.	4	2	7
	8	7	9	.	3	4	
6	2	8	4	.	9	3	2

Tenths Place

If we move one place to the right of ones place, the place value becomes $\frac{1}{10}$ of the digit and this place is known as tenths place.

For example ; Place value of 6 in 43.628 is $\frac{6}{10}$.

Hundredths Place

If we move one place to the right of tenths place, the place value becomes $\frac{1}{100}$ of the digit and this place is known as hundredths place.

For example : Place value of 2 in 43.628 is $\frac{2}{100}$.

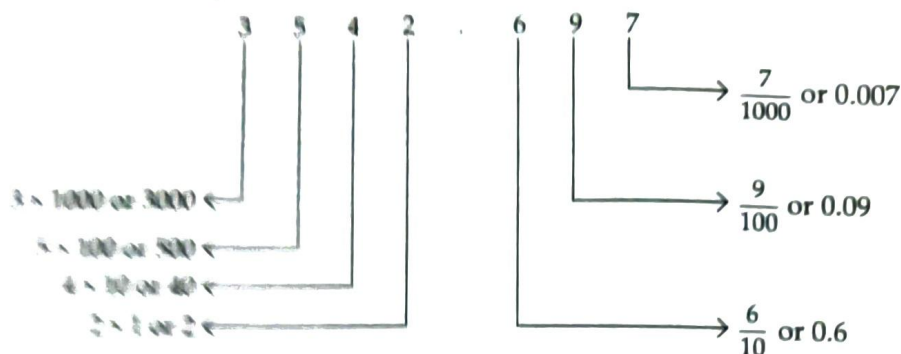
Thousandths Place

If we move one place to the right of hundredths place, the place value becomes $\frac{1}{1000}$ of the digit and this place is known as thousandths place.

For example : Place value of 8 in 43.628 is $\frac{8}{1000}$.

Expanded Form

In 3542.697, the place values are given as



3542.697 can be written as $3000 + 500 + 40 + 2 + \frac{6}{10} + \frac{9}{100} + \frac{7}{1000}$. This is known as expanded form of the decimal number.

ILLUSTRATIONS

Write the following as decimal :

- Twenty and five-tenths
- Five ones, two-tenths and four-hundredths
- Three hundred, two ones and six-hundredths
- Two ones and eight-thousandths

Soln.: (i) $20 + \frac{5}{10} = 20.5$

(ii) $5 + \frac{2}{10} + \frac{4}{100} = 5.24$

(iii) $300 + 2 + \frac{6}{100} = 302.06$

(iv) $2 + \frac{8}{1000} = 2.008$

8 Write the following as decimals :

(i) $40 + 2 + \frac{6}{10}$ (ii) $800 + 20 + \frac{8}{10} + \frac{4}{100}$

(iii) $20 + 5 + \frac{9}{10} + \frac{6}{1000}$

Soln.: (i) In $40 + 2 + \frac{6}{10}$, we have 4 tens, 2 ones and 6 tenths.

$\therefore$ Decimal is 42.6.

(ii) In $800 + 20 + \frac{8}{10} + \frac{4}{100}$, we have 8 hundreds, 2 tens, 8 tenths and 4 hundredths.

$\therefore$ Decimal is 820.84.

(iii) In $20 + 5 + \frac{9}{10} + \frac{6}{1000}$, we have 2 tens, 5 ones, 9 tenths and 6 thousandths

$\therefore$ Decimal is 25.906.

Conversion of Fractions into Decimals

Step-1 : Find a number you can multiply to the denominator of the fraction to make it 10 or 100 or 1000 etc.

Step-2 : Multiply both numerator and denominator by that number.

Step-3 : Write down just the numerator number, putting the decimal point in the correct place (one place from the right hand side for every zero in the denominator number.).

ILLUSTRATIONS

9 Convert the following fractions into decimals.

(i) $\frac{13}{5}$ (ii) $\frac{3}{4}$ (iii) $\frac{25}{2}$

Soln.: (i) $\frac{13}{5}$

Multiply the numerator and denominator by 2 to make denominator as 10.

$$\therefore \frac{13}{5} \times \frac{2}{2} = \frac{26}{10} = 2.6 \quad \therefore \frac{13}{5} = 2.6$$

(ii) $\frac{3}{4}$

Multiply the numerator and denominator by 25 to make denominator 100.

$$\therefore \frac{3}{4} \times \frac{25}{25} = \frac{75}{100} = 0.75 \quad \therefore \frac{3}{4} = 0.75$$

(iii) $\frac{25}{2}$

Multiply the numerator and denominator by 5 to make denominator as 10.

$$\therefore \frac{25}{2} \times \frac{5}{5} = \frac{125}{10} = 12.5 \quad \therefore \frac{25}{2} = 12.5$$

Conversion of Decimals into Fractions

Step-1 : Write down the decimal divided by 1, like this : $\frac{\text{Decimal}}{1}$

Step-2 : Multiply both numerator and denominator by 10 for every digit after the decimal point. (For example; if there are two digits after the decimal point, then use 100, if there are three digits then use 1000, etc.)

Step-3 : Simplify (or reduce) the fraction.

ILLUSTRATIONS

10 Convert the following decimals into fractions.

(i) 0.06 (ii) 9.645 (iii) 7.2

Soln.: (i) $0.06 = \frac{0.06}{1}$

Since, there are two digits after decimal point, so we multiply both numerator and denominator by 100.

$$\frac{0.06}{1} \times \frac{100}{100} = \frac{6}{100} = \frac{3}{50}$$

Hence, $0.06 = \frac{3}{50}$

(ii) $9.645 = \frac{9.645}{1}$

Since, there are three digits after decimal point, so we multiply both numerator and denominator by 1000.

$$\frac{9.645}{1} \times \frac{1000}{1000} = \frac{9645}{1000} = \frac{1929}{200}$$

Hence, $9.645 = \frac{1929}{200}$

(iii) $7.2 = \frac{7.2}{1}$

Since, there is one digit after decimal point, so we multiply both numerator and denominator by 10.

$$\frac{7.2}{1} \times \frac{10}{10} = \frac{72}{10} = \frac{36}{5}$$

Hence, $7.2 = \frac{36}{5}$

Use of Decimals (Units of Measurement)

Money Conversion

We know that, 100 paise = ₹ 1 $\therefore$ 1 paise = ₹ $\frac{1}{100}$ = ₹ 0.01

Weight Conversion

We know that, 1000 g = 1 kg $\therefore$ 1 g = $\frac{1}{1000}$ kg = 0.001 kg

Length Conversion

We know that, 100 cm = 1 m $\therefore$ 1 cm = $\frac{1}{100}$ m = 0.01 m

Similarly, 1000 m = 1 km $\therefore$ 1 m = $\frac{1}{1000}$ km = 0.001 km

ILLUSTRATIONS

11 Convert the following into rupees.

- (i) 560 paise (ii) 50 rupees 80 paise

Soln.: (i) 1 paise = ₹ $\frac{1}{100}$

So, 560 paise = ₹ $\frac{560}{100}$ = ₹ 5.60

(ii) 1 paise = ₹ $\frac{1}{100}$ $\therefore$ 80 paise = ₹ $\frac{80}{100}$ = ₹ 0.80

So, 50 rupees 80 paise = ₹ 50 + ₹ 0.80 = ₹ 50.80

12 Convert the following.

- (i) 2004 cm into m (ii) 8 km 42 m into km

(iii) 428 mm into cm

Soln.: (i) 1 cm = $\frac{1}{100}$ m

So, 2004 cm = $\frac{2004}{100}$ m = 20.04 m

(ii) 1 m = $\frac{1}{1000}$ km $\therefore$ 42 m = $\frac{42}{1000}$ km = 0.042 km

So, 8 km 42 m = 8 km + 0.042 km = 8.042 km

(iii) 1 mm = $\frac{1}{10}$ cm

So, 428 mm = $\frac{428}{10}$ cm = 42.8 cm

13 Convert the following into kg.

- (i) 3840 g (ii) 52 kg 50 g

Soln.: (i) 1 g = $\frac{1}{1000}$ kg

So, 3840 g = $\frac{3840}{1000}$ kg = 3.840 kg

(ii) 1 g = $\frac{1}{1000}$ kg $\therefore$ 50 g = $\frac{50}{1000}$ kg = 0.050 kg

So, 52 kg 50 g = 52 kg + 0.050 kg = 52.050 kg

Locating and Comparing Decimals

Representing Decimals on Number Line

Let us try to represent 0.8 on a number line. We know that 0.8 is greater than 0 but less than 1. So, we divide the distance between 0 and 1 into 10 equal parts on number line and name it as 0.1, 0.2 and so on till 1.

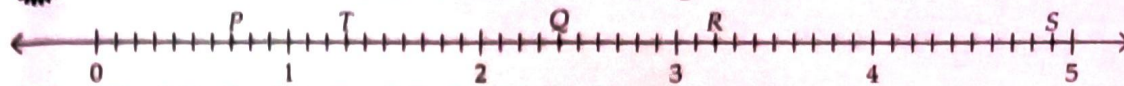


Similarly, if we had to place 0.08, we further divide distance between 0 and 0.1 into 10 equal parts and name it as 0.01, 0.02 and so on.



ILLUSTRATIONS

14 Find the values of P , Q , R , S and T on the given number line.



Soln.: $P = 0.7$, $Q = 2.4$, $R = 3.2$, $S = 4.9$ and $T = 1.3$

The Zero Dilemma in Decimals

Let us take three numbers : 1.3, 1.30 and 1.300

At first glance, they may look different but they are not.

Let's see what each number means using place value :

Decimal number	Units	Tenths	Hundredths	Thousandths
1.3	1	3	–	–
1.30	1	3	0	–
1.300	1	3	0	0

In all three numbers, the value is the same. So, $1.3 = 1.30 = 1.300$.

Thus, adding zeros on the right does not change the value of the number. These are just different ways of writing the same quantity. But zeros placed before or between digits do change the value.

For example ; $0.2 \neq 0.02 \neq 0.002$.

Here, $0.2 = 2$ tenths, $0.02 = 2$ hundredths and $0.002 = 2$ thousandths

Now, let's learn how decimal numbers are being compared.

Comparing Decimals

First, compare the whole parts to the left of the decimal point.

If they are not the same, the smaller decimal number is the one with the smaller whole part.

On the other hand, if they are the same, compare the number to the right of the decimal point. We start with tenths place and then hundredths place, etc. If one decimal has a greater digit in the tenths place, then it is larger than a decimal with smaller tenths digit. If the tenths are equal, compare the hundredths, then the thousandths etc, until we get one decimal larger than the other.

Like and Unlike Decimals

Decimals having equal number of digits to the right of the decimal point are known as like decimals and decimals having unequal number of digits to the right of the decimal point are known as unlike decimals.

Converting Unlike Decimals into Like Decimals

We can convert unlike decimals into like decimals by adding zeroes to the right of decimal part.

Now let's compare: 0.3, 0.03, and 0.003.

Decimal number	Units	Tenths	Hundredths	Thousandths
0.3	0	3	–	–
0.03	0	0	3	–
0.003	0	0	0	3

The digit 3 is in a different position in each number. So, $0.3 > 0.03 > 0.003$.

ILLUSTRATIONS

15 Which decimal is greater?

(i) 82.123 or 68.21

(ii) 0.432 or 0.598

(iii) 32.064 or 32.011

Soln.: (i) 82.123 or 68.21

Since, whole parts of both the numbers are different. Hence, $82.123 > 68.21$

(ii) 0.432 or 0.598

Since, whole parts of both the numbers are same. So, we compare tenths parts. Hence, $0.598 > 0.432$.

(iii) 32.064 or 32.011

Since, whole parts of both the numbers are same. So, we compare tenths parts. But tenths parts are also same. So, we compare hundredths parts. Hence, $32.064 > 32.011$.

16 Arrange the following in ascending order:

32.435, 32.336, 32, 32.003, 32.3

Soln.: Depending on their place values, we can arrange them in ascending order as 32, 32.003, 32.3, 32.336, 32.435.

17 Arrange the following in descending order : 65.6, 65.604, 65.064, 65, 65.664

Soln.: Depending on their place values, we can arrange them in descending order as 65.664, 65.604, 65.6, 65.064, 65.

Closest Decimals

When we have several decimal numbers and we want to know which one is closest to a given number, we compare how far each number is from that point.

Let's say we are trying to find which of the following numbers is closest to 2 : 1.89, 1.99, 2.05, 2.2

We look at how much more or less each number is from 2:

1.89 is 0.11 away from 2 ; 1.99 is 0.01 away from 2 ; 2.05 is 0.05 away from 2 and 2.2 is 0.2 away from 2

Now, we compare these differences: $0.01 < 0.05 < 0.11 < 0.2$

So, the number 1.99 is the closest to 2 because it is just 1 hundredth away.

To find the closest decimal we subtract each number from the target number (take the positive difference) and the one with the smallest difference is the closest.

ILLUSTRATIONS

18 Which of the given number is closest to 0.5?

0.45, 0.49, 0.51, 0.6

Soln.: Firstly, let us find how far each number is from 0.5

$0.5 - 0.45 = 0.05$; $0.5 - 0.49 = 0.01$; $0.51 - 0.5 = 0.01$ and $0.6 - 0.5 = 0.1$

Now, $0.01 < 0.05 < 0.1$

So, both 0.49 and 0.51 are equally close to 0.5 as both are located at a distance of 0.01 from 0.5.

Addition and Subtraction of Decimals

Addition of Decimals

To add the decimals, we proceed as follows :

Step 1 : Convert the given decimals into like decimals.

Step 2 : Write one number below the other so that the bottom decimal point is directly below and lined up with the top decimal point.

Step 3 : Then add using column addition, remembering to put the decimal point in the answer.

ILLUSTRATIONS

19 Add the following decimals.

(i) $2.237 + 0.2113 + 5.5664$

(ii) $9.176 + 1.66$

(iii) $2.3667 + 2.57 + 3.22$

Soln.: (i)

$$\begin{array}{r} 2.2370 \\ 0.2113 \\ + 5.5664 \\ \hline 8.0147 \end{array}$$

(ii)

$$\begin{array}{r} 9.176 \\ + 1.660 \\ \hline 10.836 \end{array}$$

(iii)

$$\begin{array}{r} 2.3667 \\ 2.5700 \\ + 3.2200 \\ \hline 8.1567 \end{array}$$

20 Bala had ₹ 69.24. His father gave him ₹ 100.48 and his mother gave him ₹ 50.40. How much amount he have in total?

Soln.: Amount of money Bala had = ₹ 69.24
 Amount of money his father gave = ₹ 100.48
 Amount of money his mother gave = ₹ 50.40
 Total money Bala has = ₹ 69.24 + ₹ 100.48 + ₹ 50.40
 = ₹ 220.12

Subtraction of Decimals

To subtract the decimals, we proceed as follows:

Step 1 : Convert the given decimals into like decimals.

Step 2 : Write one number below the other so that the bottom decimal point is directly below and lined up with the top decimal point.

Step 3 : Then subtract normally, remembering to put the decimal point in the answer.

ILLUSTRATIONS

21 Solve the following:

(i) Subtract 1.7609 from 6.935

(ii) Subtract 8.61 from 9.9603

(iii) Subtract 2.8536 from 3.22

Soln.: (i)

$$\begin{array}{r} 6.9350 \\ - 1.7609 \\ \hline 5.1741 \end{array}$$

(ii)

$$\begin{array}{r} 9.9603 \\ - 8.6100 \\ \hline 1.3503 \end{array}$$

(iii)

$$\begin{array}{r} 3.2200 \\ - 2.8536 \\ \hline 0.3664 \end{array}$$

22 Priya bought 5.25 kg of rice. She consumed 1.73 kg of rice. How much rice remains with her?

Soln.: Quantity of rice Priya bought = 5.25 kg
 Quantity of rice she consumed = 1.73 kg
 Quantity of rice left with her = 5.25 kg - 1.73 kg
 = 3.52 kg

Decimal Sequences

A decimal sequence is a sequence of numbers in which each term is a decimal number, and the difference between consecutive terms is either constant or follows a specific pattern. The change between terms can be an increase or a decrease.

Extending Decimal Sequences

To find the next terms, add or subtract the common difference from the previous term. This can be done mentally for small differences or patterns.

For example ; Consider the sequence: 3.5, 3.8, 4.1, 4.4, ...

On observing the sequence, we get that each time, we add 0.3 to get the next number.

As $3.5 + 0.3 = 3.8$, $3.8 + 0.3 = 4.1$ and $4.1 + 0.3 = 4.4$

So, next three terms will be 4.7, 5.0, 5.3.

Estimating Sums and Differences of Decimals

We can estimate the results of addition and subtraction of decimal numbers without calculating the exact answer. In simple terms, we find the possible range in which the sum or difference will lie.

When we add two decimal numbers, the sum will always be greater than the sum of their whole number part and always be less than 2 more than that sum.

This is a useful estimation method that helps us to predict an approximate answer, check if your final answer is reasonable and to avoid calculation errors.

The same idea can also be applied for subtraction. When we subtract two decimal numbers, the difference will be close to the difference between their whole parts.

ILLUSTRATIONS

23 Find the next two terms of sequence

12.9, 13.4, 13.9, ...

Soln.: Firstly, identify the pattern by finding the difference between terms:

$$13.4 - 12.9 = 0.5 \text{ and } 13.9 - 13.4 = 0.5$$

So, each term increases by 0.5.

Next two terms are $13.9 + 0.5 = 14.4$ and $14.4 + 0.5 = 14.9$.

24 Estimate the range of sum of $7.501 + 4.498$.

Also verify that the sum lies in the range or not.

Soln.: Whole parts = 7 and 4 and $7 + 4 = 11$

So, sum must be greater than 11.

Now, add 1 extra to each $(7+1) + (4+1) = 8+5 = 13$

So, sum must be less than 13.

Actual sum = $7.501 + 4.498 = 11.999$ which is more than 11 and less than 13. *i.e.*, lies in the desired range.

25 Subtract: $92.6 - 47.8$ and verify that the result is closer to the difference of their whole parts.

Soln.: Whole parts = 92 and 47

Difference of whole parts = $92 - 47 = 45$

So, difference of decimals should be close to 45.

Now, $92.6 - 47.8 = 44.8$, which is close to 45.

Decimals in Daily Life and History

Importance of Decimal Accuracy

Decimal points and units are crucial in calculations. Small mistakes can cause big problems. For example; A technician added 0.9 kg of chlorine instead of the required 0.09 kg to a small pool. Missing a decimal point led to 10 times more chlorine, which caused skin irritation for swimmers. Also, a student scored 18.5 out of 20, but the teacher entered it as 1.85 due to a decimal shift in the grading software. This decimal error dropped the student's average drastically and caused unnecessary stress.

Decimals can be Misleading

Decimal-like numbers sometimes do not represent true decimals:

For example; "4.5 hours past noon" is 4:30 p.m., not 4:05 or 4:50 or

Measuring 2 ft 5 inch as 2.5 ft can cause a door to be slightly wider or In cricket, 5.5 overs means 5 overs and 5 balls (since 1 over = 6 balls).

A Brief History of Decimal Notation

- Ancient Indian mathematicians, like Śhrīdharāchārya (8th century), used decimal fractions for arithmetic and algebra.
- Around 950 CE, Arab mathematician Abūl Hassan al-Uqlīdisī wrote numbers like 0.059375.
- By the 15th century, different methods separated whole numbers from fractions: vertical marks, colors, or superscripts.
- In the 16th century, John Napier and Christopher Clavius introduced the decimal point (.), while François Viète used a comma (,).

Today, some countries use a comma as a decimal separator (e.g., 1000, 5), but the decimal point is most widely used in India.

ILLUSTRATIONS

26 Convert 2.5 kg to grams.

Soln.: 1 kg = 1000 grams. In decimal numbers, 0.5 kg means half of 1000 grams, which is 500 grams, not 5 grams. So, 2.5 kg is equal to 2 kg and 500 grams. *i.e.*, $2000 + 500 = 2500$ grams

27 A digital thermometer showed a body temperature as 39.8°C , but due to a reading mistake, it was noted as 3.98°C . What is the difference between the actual and recorded temperature?

Soln.: Actual temperature = 39.8°C

Recorded temperature = 3.98°C

Difference = $39.8^{\circ}\text{C} - 3.98^{\circ}\text{C} = 35.82^{\circ}\text{C}$

CONCEPT MAP

A Tenth Part

A tenth part is one of ten equal parts of something.

A Hundredth Part

A hundredth part is one of hundred equal parts of something.

A PEEK BEYOND THE POINT

Decimal Fraction

The fractions in which the denominators are 10, 100, 1000, etc..., are known as decimal fractions.

Use of Decimals

Length, weight, money

Addition and Subtraction of Decimals

Addition of Decimals

To add the decimals, we proceed as follows:
Step 1 : Convert the given decimals into like decimals.

Step 2 : Write one number below the other so that the bottom decimal point is directly below and lined up with the top decimal point.

Step 3 : Then add using column addition, remembering to put the decimal point in the answer.

Subtraction of Decimals

To subtract the decimals, we proceed as follows:

Step 1 : Convert the given decimals into like decimals.

Step 2 : Write one number below the other so that the bottom decimal point is directly below and lined up with the top decimal point.

Step 3 : Then subtract normally, remembering to put the decimal point in the answer.

Decimal Sequences

A decimal sequence is a sequence of numbers in which each term is a decimal number, and the difference between consecutive terms is either constant or follows a specific pattern. The change between terms can be an increase or a decrease.

Solved Examples

1. Which is greater?

(i) $4\frac{3}{10}$ or $\frac{43}{10}$ (ii) $6\frac{2}{10}$ or $6\frac{29}{100}$

Soln.: (i) Firstly let us convert the mixed number into tenths

$$4\frac{3}{10} = 4 + \frac{3}{10} = \frac{40}{10} + \frac{3}{10} = \frac{43}{10}$$

Now, let us compare $4\frac{3}{10}$ and $\frac{43}{10}$.

We just found that $4\frac{3}{10} = \frac{43}{10}$ i.e., both are equal.

(ii) $6\frac{2}{10} = \frac{60}{10} + \frac{2}{10} = \frac{62}{10} = \frac{62 \times 10}{10 \times 10} = \frac{620}{100}$ and

$$6\frac{29}{100} = \frac{600}{100} + \frac{29}{100} = \frac{629}{100}$$

Now, $629 > 620$ so, $\frac{629}{100} > \frac{620}{100}$ i.e., $6\frac{29}{100} > 6\frac{2}{10}$

2. Convert: $4\frac{2}{10}\frac{5}{100}$ into a single fraction with denominator 100.

Soln.: We have, $4 = \frac{400}{100}$ and $\frac{2}{10} = \frac{20}{100}$

$$\text{So, } 4\frac{2}{10}\frac{5}{100} = 4 + \frac{2}{10} + \frac{5}{100} = \frac{400}{100} + \frac{20}{100} + \frac{5}{100} = \frac{425}{100}$$

3. Arrange in ascending order:

$$4\frac{4}{10}, 4\frac{23}{100}, 4\frac{105}{100}, 4\frac{5}{10}$$

Soln.: $4\frac{4}{10} = \frac{400}{100} + \frac{40}{100} = \frac{440}{100}$,

$$4\frac{23}{100} = \frac{400}{100} + \frac{23}{100} = \frac{423}{100}$$

$$4\frac{105}{100} = \frac{400}{100} + \frac{105}{100} = \frac{505}{100}$$

and $4\frac{5}{10} = \frac{400}{100} + \frac{50}{100} = \frac{450}{100}$

Now, $423 < 440 < 450 < 505$.

$$\text{So, } \frac{423}{100} < \frac{440}{100} < \frac{450}{100} < \frac{505}{100}$$

$$\text{i.e. } 4\frac{23}{100} < 4\frac{4}{10} < 4\frac{5}{10} < 4\frac{105}{100}$$

4. A rope is $5\frac{3}{10}$ meters long. Another rope is $4\frac{9}{10}$ meters long. How much longer is the first rope than the second one?

Soln.: Length of 1st rope = $5\frac{3}{10}$ meters

Length of 2nd rope = $4\frac{9}{10}$ meters

$$\begin{aligned} \text{Difference in length of rope} &= 5\frac{3}{10} - 4\frac{9}{10} \\ &= (5 - 4) + \left(\frac{3}{10} - \frac{9}{10}\right) = 1 - \frac{6}{10} = \frac{10}{10} - \frac{6}{10} = \frac{4}{10} \end{aligned}$$

So, first rope is $\frac{4}{10} = 0.4$ meters longer than the second one.

5. Divya bought $3\frac{7}{10}$ meters of cloth from one shop and $2\frac{45}{100}$ meters from another. Then, she stitched both together with a waste loss of $\frac{1}{10}$ meter during cutting. What is the final usable length of fabric Divya has?

Soln.: Total length of both clothes = $3\frac{7}{10} + 2\frac{45}{100}$

$$= (3 + 2) + \left(\frac{7}{10} + \frac{45}{100}\right) = 5 + \left(\frac{70}{100} + \frac{45}{100}\right) = 5 + \frac{115}{100}$$

$$= 5 + \frac{100}{100} + \frac{15}{100} = 6\frac{15}{100} \text{ meters}$$

$$\begin{aligned} \text{Now, loss of cloth during cutting} &= \frac{1}{10} \text{ meter} \\ &= \frac{10}{100} \text{ meter} \end{aligned}$$

$$\text{Length of usable fabric} = 6\frac{15}{100} - \frac{10}{100} = 6\frac{5}{100} \text{ meters}$$

6. A tank was filled with $12\frac{38}{100}$ litres of water in the morning and then $7\frac{8}{10}$ litres in the evening. The total capacity of the tank is 20 litres. Did the tank overflow, or how much more water can be added?

Soln.: Quantity of water poured in tank

$$= 12 \frac{38}{100} + 7 \frac{8}{10} = (12 + 7) + \left(\frac{38}{100} + \frac{8}{10} \right)$$

$$= 19 + \left(\frac{38}{100} + \frac{80}{100} \right) = 19 \frac{118}{100} = 20 \frac{18}{100}$$

But capacity of water is 20 litres and water poured in tank is $20 \frac{18}{100}$ litres which is greater than 20 litres so, the tank overflow by $\frac{18}{100}$ litres.

7. A chocolate bar is $3 \frac{8}{10} \frac{15}{100}$ units long.

Two pieces of $1 \frac{3}{10}$ units each are cut from it.

What is the remaining length?

Soln.: Total length of two pieces cut from

$$\text{chocolate bar} = 1 \frac{3}{10} + 1 \frac{3}{10} = 2 \frac{6}{10} \text{ units}$$

$$\text{Length of chocolate bar} = 3 \frac{8}{10} \frac{15}{100} \text{ units}$$

$$\text{remaining length} = 3 \frac{8}{10} \frac{15}{100} - 2 \frac{6}{10}$$

$$= (3 - 2) + \left(\frac{80}{100} + \frac{15}{100} - \frac{60}{100} \right) = 1 \frac{35}{100} \text{ units}$$

Length of remaining bar is $1 \frac{35}{100}$ units.

8. Write down the measurements for the lengths shown and describe each one in words.

(i)



10. Write the following numbers in the Place value chart (table).

(i) 142.76

(ii) 32.345

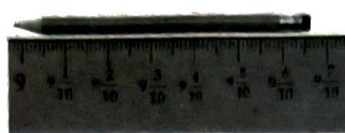
(iii) 1763.42

Soln.:

	Number	Thousands	Hundreds	Tens	Ones	Decimal	Tenths	Hundredths	Thousandths
(i)	142.76		1	4	2	.	7	6	
(ii)	32.345			3	2	.	3	4	5
(iii)	1763.42	1	7	6	3	.	4	2	

11. Sundaram bought a toothpaste for ₹ 18.75, soap for ₹ 6 and shoe polish for ₹ 12.50. He gave a fifty rupee note to the shopkeeper. Find the money he got back.

(ii)



Soln.: (i) The length of the object is measured

$$\text{as } 7 \frac{3}{10} \frac{2}{100} - 6 \frac{6}{10} \frac{5}{100} = (7 - 6) + \left(\frac{3}{10} - \frac{6}{10} \right) + \left(\frac{2}{100} - \frac{5}{100} \right)$$

$$= 1 - \frac{3}{10} \frac{3}{100} = \frac{100}{100} - \frac{30}{100} - \frac{3}{100} = \frac{67}{100} = \frac{6}{10} \frac{7}{100}$$

and read as six-tenths and seven-hundredths.

Also, $\frac{6}{10} \frac{7}{100} = \frac{67}{100}$ read as sixty seven-hundredths.

(ii) The length of the object is measured as

$$9 \frac{6}{10} \frac{6}{100} - 8 \frac{9}{10} \frac{8}{100} = (9 - 8) + \left(\frac{6}{10} - \frac{9}{10} \right) + \left(\frac{6}{100} - \frac{8}{100} \right)$$

$$= 1 - \frac{3}{10} - \frac{2}{100} = \frac{100}{100} - \frac{30}{100} - \frac{2}{100} = \frac{68}{100} = \frac{6}{10} \frac{8}{100}$$

and read as six-tenths and eight-hundredths.

Also, $\frac{6}{10} \frac{8}{100} = \frac{68}{100}$ read as sixty eight-hundredths.

9. Write the decimal number that has 4 ones, 5 tenths and 2 hundredths.

Soln.: The number that has 4 ones, 5 tenths and 2 hundredths

$$= 4 + \frac{5}{10} + \frac{2}{100} = 4 + 0.5 + 0.02 = 4.52$$

12. Write the decimal numbers represented by the points A, B and C on the number line.



Soln.: $A = 1.1$, $B = 2.7$, $C = 3.5$

13. Compare the following.

- (i) 63.84 and 57.98 (ii) 27.04 and 27.10
(iii) 82.438 and 82.434 (iv) 91.109 and 91.121

Soln.: (i) $63.84 > 57.98$ (ii) $27.04 < 27.10$
(iii) $82.438 > 82.434$ (iv) $91.109 < 91.121$

14. Write each of the following as decimals.

(i) $300 + 60 + 9 + \frac{5}{10} + \frac{8}{100}$

(ii) $700 + 9 + \frac{2}{10} + \frac{6}{1000}$

(iii) $\frac{8}{10} + \frac{5}{100} + \frac{4}{1000}$ (iv) $20 + 4 + \frac{1}{100}$

Soln.: (i) $300 + 60 + 9 + \frac{5}{10} + \frac{8}{100} = 369.58$

(ii) $700 + 9 + \frac{2}{10} + \frac{6}{1000} = 709.206$

(iii) $\frac{8}{10} + \frac{5}{100} + \frac{4}{1000} = 0.854$

(iv) $20 + 4 + \frac{1}{100} = 24.01$

15. Using decimals, express:

- (i) 10 cm 6 mm as cm
(ii) 108 cm as m
(iii) ₹ 5 and 40 paise as rupees
(iv) ₹ 50 and 18 paise as rupees

Soln.: (i) $10 \text{ cm } 6 \text{ mm} = 10 \text{ cm} + \frac{6}{10} \text{ cm}$
 $= 10 \text{ cm} + 0.6 \text{ cm} = 10.6 \text{ cm}$

(ii) $108 \text{ cm} = \frac{108}{100} \text{ m} = 1.08 \text{ m}$

(iii) $₹ 5 + 40 \text{ paise} = ₹ 5 + ₹ \frac{40}{100}$
 $= ₹ 5 + ₹ 0.40 = ₹ 5.40$

(iv) $₹ 50 + 18 \text{ paise} = ₹ 50 + ₹ \frac{18}{100} = ₹ 50 + ₹ 0.18 = ₹ 50.18$

16. Shweta travelled 12 km 220 m by bus, 8 km 485 m by car and the rest 760 m she walked. How much distance did she travel in all?

Soln.: Distance travelled by bus = 12 km 220 m
 $= 12.220 \text{ km}$

Distance travelled by car = 8 km 485 m = 8.485 km

Distance travelled by walking = 760 m = 0.760 km
Therefore, total distance travelled
 $= 12.220 \text{ km} + 8.485 \text{ km} + 0.760 \text{ km} = 21.465 \text{ km}$
Hence, Shweta travelled 21.465 km in all.

17. Write the following decimals in expanded form.

- (i) 2.75 (ii) 542.344 (iii) 76.08 (iv) 1563.704

Soln.: (i) $2.75 = 2 + \frac{7}{10} + \frac{5}{100}$

(ii) $542.344 = 500 + 40 + 2 + \frac{3}{10} + \frac{4}{100} + \frac{4}{1000}$

(iii) $76.08 = 70 + 6 + \frac{8}{100}$

(iv) $1563.704 = 1000 + 500 + 60 + 3 + \frac{7}{10} + \frac{4}{1000}$

18. Tanuj walked 8.62 km on Monday, 7.05 km on Tuesday and some distance on Wednesday. If he walked 21015 m in three days, how much distance did he walk on Wednesday?

Soln.: Distance walked by Tanuj on Monday
 $= 8.62 \text{ km}$

Distance walked by Tanuj on Tuesday = 7.05 km

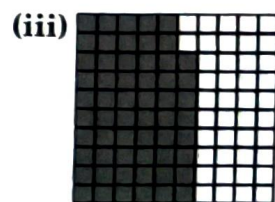
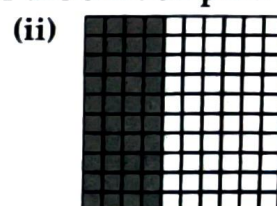
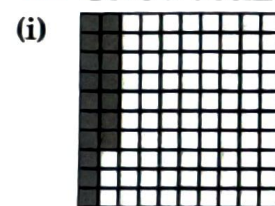
Total Distance walked by Tanuj on Monday and Tuesday = 8.62 km + 7.05 km
 $= 15.67 \text{ km} = 15.670 \text{ km}$

Distance walked by Tanuj on Monday, Tuesday and Wednesday = 21015 m

$= \frac{21015}{1000} \text{ km} = 21.015 \text{ km}$

$\therefore$ Distance walked by Tanuj on Wednesday
 $= 21.015 \text{ km} - 15.670 \text{ km} = 5.345 \text{ km}$

19. Give a decimal for the shaded part.



Soln.: (i) 17 parts are shaded out of 100 parts
i.e., $\frac{17}{100} = 0.17$

(ii) 40 parts are shaded out of 100 parts

$$\text{i.e., } \frac{40}{100} = 0.40$$

(iii) 58 parts are shaded out of 100 parts

$$\text{i.e., } \frac{58}{100} = 0.58$$

20. Write the place value of 6 in the following decimal numbers.

(i) 28.63 (ii) 15.16 (iii) 8.036

Soln.: (i) Place value of 6 in 28.63 is $\frac{6}{10}$.

(ii) Place value of 6 in 15.16 is $\frac{6}{100}$.

(iii) Place value of 6 in 8.036 is $\frac{6}{1000}$.

21. A number is 61.172 less than 75. Find the sum of number with 40.69. By how much the sum is more than 50?

Soln.: Number which is 61.172 less than 75
 $= 75 - 61.172 = 13.828$

Now, sum of 13.828 and 40.69 $= 13.828 + 40.690$
 $= 54.518$

$\therefore$ Required difference $= 54.518 - 50 = 4.518$

Thus, 54.518 is 4.518 more than 50.

22. Estimate the range of the sum: $4.7 + 3.9$

Soln.: From the given numbers we have, whole part of 4.7 = 4 and whole part of 3.9 = 3

Now add just the whole parts : $4 + 3 = 7$.

So, the sum will be greater than 7.

Now, add 1 extra to each whole part $(4 + 1) + (3 + 1)$
 $= 5 + 4 = 9$

So, sum must be less than 9.

Actual sum $= 4.7 + 3.9 = 8.6$ lies between 7 and 9.

NCERT Section

Figure it Out

(Page 58)

1. Find the sums and differences :

(a) $\frac{3}{10} + 3\frac{4}{100}$ (b) $9\frac{5}{10}\frac{7}{100} + 2\frac{1}{10}\frac{3}{100}$

(c) $15\frac{6}{10}\frac{4}{100} + 14\frac{3}{10}\frac{6}{100}$

(d) $7\frac{7}{100} - 4\frac{4}{100}$ (e) $8\frac{6}{100} - 5\frac{3}{100}$

(f) $12\frac{6}{100}\frac{2}{100} - \frac{9}{10}\frac{9}{100}$

Figure it Out

(Page 75)

1. Find the sums

- (a) $5.3 + 2.6$ (b) $18 + 8.8$
(c) $2.15 + 5.26$ (d) $9.01 + 9.10$
(e) $29.19 + 9.91$ (f) $0.934 + 0.6$
(g) $0.75 + 0.03$ (h) $6.236 + 0.487$

2. Find the differences.

- (a) $5.6 - 2.3$ (b) $18 - 8.8$
(c) $10.4 - 4.5$ (d) $17 - 16.198$

(e) $17 - 0.05$

(g) $9.9 - 9.09$

(f) $34.505 - 18.1$

(h) $6.236 - 0.487$

Figure it Out

(Page 78-80)

1. Convert the following fractions into decimals:

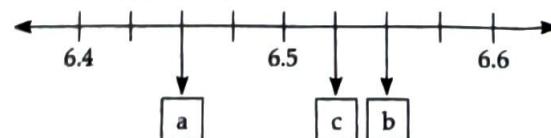
(a) $\frac{5}{100}$ (b) $\frac{16}{1000}$

(c) $\frac{12}{10}$ (d) $\frac{254}{1000}$

2. Convert the following decimals into a sum of tenths, hundredths, and thousandths:

- (a) 0.34 (b) 1.02
(c) 0.8 (d) 0.362

3. What decimal number does each letter represent in the number line below ?



4. Arrange the following quantities in descending order:

- (a) 11.01, 1.011, 1.101, 11.10, 1.01

- (b) 2.567, 2.675, 2.768, 2.499, 2.698
 (c) 4.678 g, 4.595 g, 4.600 g, 4.656 g, 4.666 g
 (d) 33.13 m, 33.31 m, 33.133 m, 33.331 m, 33.313 m

5. Using the digits 1, 4, 0, 8, and 6 make :

- (a) The decimal number closest to 30
 (b) The smallest possible decimal number between 100 and 1000.

6. Will a decimal number with more digits be greater than a decimal number with fewer digits?

7. Mahi purchases 0.25 kg of beans, 0.3 kg of carrots, 0.5 kg of potatoes, 0.2 kg of capsicums, and 0.05 kg of ginger. Calculate the total weight of the items she bought.

8. Pinto supplies 3.79 L, 4.2 L, and 4.25 L of milk to a milk dairy in the first three days. In 6 days, he supplies 25 litres of milk. Find the total quantity of milk supplied to the dairy in the last three days.

9. Tinku weighed 35.75 kg in January and 34.50 kg in February. Has he gained or lost weight? How much is the change?

10. Extend the pattern: 5.5, 6.4, 6.39, 7.29, 7.28, 6.18, 6.17, _____

Note : In this questions the correct pattern is :

5.5, 6.4, 6.39, 7.29, 7.28, 8.18, 8.17

11. How many millimetres make 1 kilometre?

12. Indian Railways offers optional travel insurance for passengers who book e-tickets. It

costs 45 paise per passenger. If 1 lakh people opt for insurance in a day, what is the total insurance fee paid?

13. Which is greater?

(a) $\frac{10}{1000}$ or $\frac{1}{10}$?

(b) One-hundredth or 90 thousandths?

(c) One-thousandth or 90 hundredths?

14. Write the decimal forms of the quantities mentioned (an example is given):

(a) 87 ones, 5 tenths and 60 hundredths = 88.10

(b) 12 tens and 12 tenths

(c) 10 tens, 10 ones, 10 tenths, and 10 hundredths

(d) 25 tens, 25 ones, 25 tenths, and 25 hundredths

15. Using each digit 0 – 9 not more than once, fill the boxes below so that the sum is closest to 10.5:

$$\begin{array}{r} \square . \square \square \square \\ + \square . \square \square \square \\ \hline \end{array}$$

16. Write the following fractions in decimal form:

(a) $\frac{1}{2}$ (b) $\frac{3}{2}$ (c) $\frac{1}{4}$ (d) $\frac{3}{4}$

(e) $\frac{1}{5}$ (f) $\frac{4}{5}$

Exercise

Multiple Choice Questions

LEVEL - 1

Decimals

1. Which of the following pairs will give a sum less than 7?

(a) $4\frac{35}{100}$ and $2\frac{55}{100}$ (b) $3\frac{75}{100}$ and $3\frac{20}{100}$

(c) $5\frac{4}{10}$ and $1\frac{1}{10}$ (d) All of these

(NCERT Page 52)

2. How many hundredths are there in 4 tenths?

(a) 400 (b) 40

(c) $\frac{1}{40}$ (d) 4 (Page 53) 

3. In 16.532, the digit 3 is in which place?

- (a) Tenths
 (b) Hundredths
 (c) Hundreds
 (d) Thousands

(Page 53) 

4. What is the total of $2\frac{8}{10}$, $1\frac{45}{100}$ and $3\frac{5}{10}$?
- (a) $7\frac{35}{100}$ (b) $7\frac{25}{100}$
 (c) $7\frac{75}{100}$ (d) $6\frac{55}{100}$ (Page 56)

5. Find the value of $10 + 0 + \frac{9}{100} + \frac{5}{1000}$.
- (a) 10.905 (b) 10.95
 (c) 10.095 (d) 1.095 (Page 62)

6. The expanded form of 28.001 is _____.
- (a) $20 + 8 + \frac{1}{10}$ (b) $20 + 8 + \frac{1}{1000}$
 (c) $20 + 8 + \frac{1}{100}$ (d) $20 + \frac{8}{10} + \frac{1}{100}$ (Page 62)

7. Write the following as decimal.
- $$200 + 10 + 5 + \frac{6}{10}$$
- (a) 205.6 (b) 215.06
 (c) 205.16 (d) 215.6 (Page 62) **U**

8. $60 + 6 + \frac{4}{10} + \frac{8}{100}$ in decimal form can be written as
- (a) 606.48 (b) 66.048
 (c) 606.048 (d) 66.48 (Page 62)

9. 15 mm = _____ m
- (a) 0.15 (b) 0.015
 (c) 1.5 (d) 1.05 (Page 64) **An**

10. 2 m is equal to
- (a) 0.2 km (b) 0.02 km
 (c) 0.002 km (d) 0.0002 km (Page 64) **R**

11. 5 kg 3 g is expressed in kg using decimal as
- (a) 3.005 kg (b) 5.003 kg
 (c) 5.3 kg (d) 5.03 kg (Page 67) **R**

12. The place value of '7' in 32.507 is
- (a) 7 hundredths (b) 7 tenths
 (c) 7 thousandths (d) 7 thousands

13. The value of $(1 + 0.1 + 0.01 + 0.001)$ is
- (a) 1.001 (b) 1.111
 (c) 0.111 (d) 1.011 **Ev**

14. Which of the following decimal numbers are represented by points K and L on the given number line?



- (a) 0.5, 1.5 (b) 1.5, 1.5
 (c) 0.12, 0.5 (d) 1.5, 2.5 **U**

Comparison, Addition and Subtraction of Decimals

15. Which of the following represents the same quantity?
- (a) 1.4 (b) 1.40
 (c) 1.400 (d) All of these (Page 69)
16. Which of the following sums will be greater than 10?
- (a) $4.2 + 5.3$ (b) $6.3 + 3.9$
 (c) $2.5 + 7.4$ (d) (b) and (c) both (Page 69)

17. The largest among 54.32, 54.35, 54.23, and 5.23 is
- (a) 54.32 (b) 54.23
 (c) 54.35 (d) 5.23 (Page 70)

18. Which is closest to 5?
- (a) 4.85 (b) 5.05
 (c) 5.2 (d) 5.99 (Page 71)

19. The sum of $609.20 + 16.3 + 42$ is
- (a) 660.5 (b) 667.5
 (c) 669.5 (d) 670.5 (Page 71)

20. Anusha has three pieces of ribbons. One piece is 43.75 m long, another 82.6 m long and the third is 63.25 m long. What is the total length of the three pieces of ribbon?
- (a) 189.60 m (b) 204.40 m
 (c) 174.60 m (d) 196.40 m (Page 74) **B**

21. A man had a hundred rupee note. He bought vegetables for ₹ 26.75 and fruits for ₹ 25.40. How much money he has left?
- (a) ₹ 47.80 (b) ₹ 52.40
 (c) ₹ 73.25 (d) ₹ 37.80 (Page 74) **B**

22. $43.266 - 24.557 + 34.970$ is simplified as
- (a) 53.769 (b) 53.679
 (c) 753.679 (d) 102.793 (Page 75)

23. Simplify the following:
 $43.16 + 493.28 - 507.34$
- (a) 29.5 (b) 29.1
 (c) 417.54 (d) 143.78 (Page 75)

24. Simplify the following:

$$2.36 - 3.24 + 4.57$$

- (a) 3.96 (b) 10.17
(c) 3.69 (d) 1.03

(Page 74)

25. Determine the next two terms of sequence:

1.1, 0.9, 0.7, 0.5, __, __

- (a) 0.3 and 0.1
(b) 1.3 and 1.5
(c) 0.2 and 0.1
(d) None of these

(Page 75)

Ap

26. Estimate the sum of 7.8 and 2.1.

- (a) between 8 and 10
(b) between 9 and 11
(c) between 10 and 12
(d) between 6 and 9

(Page 76)

An

27. Which statement is true?

- (a) $5.69 < 5.6$ (b) $10.61 > 10.602$
(c) $11.112 < 2.112$ (d) $12.55 > 12.555$

U

28. Which decimal number is greater than $\frac{3}{4}$?

- (a) 0.85 (b) 0.5 (c) 0.69 (d) 0.75

29. What is the sum of 6 mm, 60 m and 60 cm?

- (a) 6600.6 cm (b) 6000.6 cm
(c) 6060.6 cm (d) 6666 cm

Ap

30. Subtracting 8 km 73 m from 15 km 160 m, we get _____.

- (a) 7.078 km (b) 0.087 km
(c) 23.233 km (d) 7.087 km

31. Subtract 1.412 km from 3.69 km.

- (a) 1.728 km (b) 2.278 km
(c) 2.728 km (d) 1.278 km

32. Add : 9.7, 13.85, 126.05 and 8.796.

- (a) 22.883 (b) 158.396
(c) 1583.96 (d) 228.83

Ev

33. $32.1068 - 2.009$ is equal to

- (a) 30.0978 (b) 300.978
(c) 30.00978 (d) 30.978

LEVEL - 2

34. Arnav baked a cake that was $4\frac{7}{10}\frac{12}{100}$ units

in length. He cut out a piece of $2\frac{5}{10}\frac{62}{100}$ units. How much of the cake remains?

- (a) $1\frac{2}{10}\frac{50}{100}$ (b) $2\frac{2}{10}\frac{30}{100}$

(c) $1\frac{2}{10}\frac{40}{100}$

(d) $2\frac{60}{100}$

Ap

35. Which of the following statements is not true about the sum of $5\frac{9}{10}\frac{30}{100} + \frac{32}{10} + 3\frac{2}{10}\frac{75}{100}$?

- (a) The sum is more than 9
(b) The sum is equal to $9\frac{12}{10}\frac{5}{100}$
(c) The tenths and hundredths together exceed 100
(d) The sum can be written as $10\frac{1}{10}\frac{5}{100}$

U

36. Which of the following set of decimals are arranged in ascending order?

- (a) 2.910, 2.019, 2.109, 2.119
(b) 2.910, 2.119, 2.109, 2.019
(c) 2.019, 2.109, 2.119, 2.910
(d) 2.119, 2.109, 2.019, 2.910

An

37. Pushkar purchased a chocolate for ₹ 12.50, a pastry for ₹ 5.75 and a packet of chips for ₹ 8.38. How much money did he spend in all?

- (a) ₹ 27.1 (b) ₹ 26.63
(c) ₹ 28.63 (d) ₹ 20

38. Find the value of

- (i) $17.432 - 12.876 + 82.49 - 51.905$
(ii) $39.2186 + 23.543 - 82.432 + 90.625$
(iii) $69.3875 - 37.483 - 28.515 + 38.75$

- | (i) | (ii) | (iii) |
|------------|---------|---------|
| (a) 42.645 | 80.1955 | 42.1395 |
| (b) 35.141 | 80.1955 | 35.4325 |
| (c) 42.645 | 70.9546 | 35.4325 |
| (d) 35.141 | 70.9546 | 42.1395 |

Ev

39. The corner store sold 129.91 litres of milk on Thursday and 11.32 litres more than this quantity on Friday. The following day, 18.94 litres less milk was sold than on Friday. How many litres of milk did they sell on Saturday?

- (a) 160.17 litres (b) 137.53 litres
(c) 122.29 litres (d) 99.65 litres

40. Which of the following is the correct?

- (a) $2.2 < 2.02 < 2.002 < 2.222$
(b) $2.002 < 2.02 < 2.2 < 2.222$
(c) $2.02 < 2.22 < 2.002 < 2.222$
(d) None of these

U

41. Simplify the following :
 $75.3 - 104.645 + 178.96 - 47.9$
 (a) 95.715 (b) 86.475
 (c) 101.715 (d) 100.475 **Ev**
42. By how much should 23.745 be increased to get 35?
 (a) 11.255 (b) 10.215
 (c) 11.612 (d) 12.155 **An**
43. What is the sum of 5 cm, 50 m and 50 cm?
 (a) 50.55 m (b) 50.055 m
 (c) 55.05 m (d) 55.055 m **Ap**
44. Which of the following is correct?
 (a) $0.088 > 0.18$ (b) $0.4 = 0.04$
 (c) $4.72 > 4.702$ (d) $2.8 > 2.800$
45. 8 kg 70 g - 0.526 kg is equal to
 (a) 7.544 kg (b) 7.455 kg
 (c) 8 kg (d) 75.44 g
46. The decimal for 3 hundreds + 8 hundredths is
 (a) 3.8 (b) 300.8
 (c) 300.08 (d) 300.008 **U**
47. Mr. Sharma spent ₹ 1287.90 on clothings, ₹ 389.96 on travelling and ₹ 1280.71 for hotel accomodation. If he has ₹ 5800 left with him, how much money did he have in the beginning?
 (a) ₹ 5875.75 (b) ₹ 8758.57
 (c) ₹ 7858.58 (d) ₹ 8578.57 **Ev**
48. A mountaineer climbs 67.32 metres from the base camp situated at an altitude of 2307.58 metres. He climbs further 75.86 metres. What altitude does he reach after this?
 (a) 2450.76 m (b) 2050.67 m
 (c) 1450.76 m (d) 2105.76 m
49. Mishika went to the market with ₹ 4000. She purchased one top, one toy and skates costing ₹ 578.75, ₹ 309 and ₹ 1862.85 respectively. How much money is left with her?
 (a) ₹ 2249.40 (b) ₹ 2750.60
 (c) ₹ 1249.40 (d) ₹ 1750.60
50. $15.8 - 6.73$ is equal to
 (a) 8.07 (b) 9.07 (c) 9.13 (d) 9.25
51. 7 litres 6 millilitres = _____ litres
 (a) 7.06 (b) 7.6

- (c) 7.006 (d) None of these

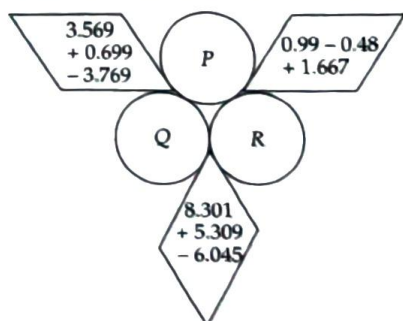
52. Rohan, Mahi, Zeba and Jasmine appeared in a Olympiad exam and got marks 200.40, 200.75, 200.25 and 200.30 respectively out of 500. Who got the highest marks?
 (a) Rohan (b) Mahi
 (c) Jasmine (d) Zeba **U**

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

53. The simplification of $2.002 + 7.9 + \{2.8 - 6.3 + (3.6 - 1.5) + 15.6\}$ gives
 (a) 2.002 (b) 4.2845
 (c) 40.843 (d) 24.102 **Ev**
54. Kiara weighs 34.5 kg. Her father weighs 69 kg more than Kiara's weight. What is their total weight?
 (a) 69 kg (b) 103.50 kg
 (c) 138 kg (d) 172.50 kg
55. Garima was shopping for a new hand bag. The one she wanted to buy cost ₹ 428.98. The sales person informed her that the same hand bag would be on sale in the following week for ₹ 399.99. How much money would she save by waiting until the hand bag went on sale?
 (a) ₹ 28.99 (b) ₹ 29.01
 (c) ₹ 128.99 (d) ₹ 39.09 **An**
56. A drum has 705 litres of oil. Two containers of capacity 135.75 litres and 253.50 litres are filled with oil from the drum. Find the quantity of oil left in the drum.
 (a) 720 litres (b) 300 litres
 (c) 215.75 litres (d) 315.75 litres
57. Read the statements carefully and state T for true and F for false.
- I. The place value of a digit at tenths place is 0.1 times the place value of same digit at the hundredths place.
- II. $\frac{0.03}{0.3} + \frac{3.3}{0.03} = 110.1$
- III. Five and seven hundredths is written as 5.07
- | I | II | III |
|-------|----|-----|
| (a) F | T | T |
| (b) F | F | F |
| (c) T | F | F |
| (d) F | T | F |

58. Simplify the given expression :
 $5 + 0.6 + 0.03 + 0.004 + (9 \times 100) + (2 \times 10)$
 (a) 529.364 (b) 259.634
 (c) 925.634 (d) 925.436 **Ev**
59. Compare and fill in the box with $<$, $>$ or $=$.
 $18.8 + 3\frac{1}{5} + 7\frac{3}{5}$ $16.2 + 4\frac{7}{25} + 3\frac{1}{8}$
 (a) = (b) $>$
 (c) $<$
 (d) Can't be determined **An**
60. A line segment measuring 28.254 cm is divided into three parts. If lengths of two line segments are 11.678 cm and 7.158 cm, then find the measure of third part.
 (a) 4.112 cm (b) 5.646 cm
 (c) 9.264 cm (d) 9.418 cm
61. Arrange the following numbers in descending order.
 0.9898, 0.8899, 0.9889, 0.8989
 (a) 0.9898, 0.9889, 0.8989, 0.8899
 (b) 0.8899, 0.8989, 0.9898, 0.9889
 (c) 0.8899, 0.8989, 0.9889, 0.9898
 (d) 0.8989, 0.8899, 0.9889, 0.8989 **Ap**
62. If the sum of two numbers is 15.96 and one number is 3.58, find the other number.
 (a) 12.38 (b) 19.54 (c) 13.82 (d) 15.94
63. Chinmay purchased 10 kg rice for ₹ 150, 15 kg 40 g sugar for ₹ 300.80 and 5 kg 60 g salt for ₹ 45.54. He gave ₹ 1000 to the shopkeeper. How much money did he get back?
 (a) ₹ 234.94 (b) ₹ 503.66
 (c) ₹ 496.34 (d) ₹ 743.99 **Ap**
64. Find the values of circles P, Q and R respectively, such that the sum of the numbers in two quadrilaterals equals to the value of the circle in between them.



- (a) 2.676, 8.064, 9.742
 (b) 7.334, 6.442, 2.332
 (c) 9.742, 2.676, 8.064
 (d) 8.064, 9.742, 8.999 **Ev**

65. A courier man covers 35 m distance to deliver a product to the customer. He travels 0.028 km by bicycle and the rest on foot. What distance does he cover on foot?
 (a) 5 m (b) 7 m (c) 8 m (d) 10 m
66. The difference between place values of digit 5 in 456.385 is _____.
 (a) 49995 (b) 499.95 (c) 49.995 (d) 49.999
67. Sanya bought two similar pens for ₹ 13.60 each and a notebook for ₹ 46.35. She gave the cashier a ₹ 100 note. How much money did she get back?
 (a) ₹ 26.45 (b) ₹ 26.76
 (c) ₹ 25.87 (d) ₹ 28.46
68. The distance between city A and city B is 217.352 km. A man travelling from city A to city B covers a distance of 108.178 km. How much more distance he has to cover?
 (a) 111.226 km (b) 109.174 km
 (c) 109.147 km (d) 325.53 km **Ap**
69. Beena bought 12 kg of sugar. After filling 11 containers of the same size, she had 3.2 kg left. How much sugar was there in 11 containers?
 (a) 0.5 kg (b) 0.6 kg (c) 0.7 kg (d) 8.8 kg
70. A mysterious decimal sequence follows this rule:
 First 3 terms are: 0.48, 0.6, 0.84
 Next 3 terms are: 1.2, 1.68, 2.28
 If this pattern continues, what will be the 8th term in the sequence?
 (a) 3.24 (b) 3.12
 (c) 3.48 (d) None of these



Match the Following

In this section each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which one is correct.

1. Match the following :

List-I	List-II
(P) 4 full units and 4 tenths	1. $4\frac{14}{100}$
(Q) Four and fourteen-hundredths	2. $4 + \frac{4}{10}$
(R) Four hundred-hundredths	3. 4
(S) 4 tenths + 5 hundredths	4. 0.45

- (a) P - 1, Q - 3, R - 4, S - 2
 (b) P - 1, Q - 3, R - 2, S - 4
 (c) P - 2, Q - 1, R - 3, S - 4
 (d) P - 2, Q - 1, R - 4, S - 3

2. Match the following

List-I

List-II

(P) $2\frac{7}{10} + 1\frac{4}{10}$

1. $4\frac{10}{100}$

(Q) $6\frac{1}{10} - 2\frac{7}{10}$

2. $5\frac{55}{100}$

(R) $2\frac{4}{10} \frac{70}{100} + 1\frac{145}{100}$

3. $3\frac{4}{10}$

(S) $6\frac{23}{100} - 2\frac{58}{100}$

4. $3\frac{65}{100}$

- (a) P - 1, Q - 3, R - 4, S - 2
 (b) P - 1, Q - 3, R - 2, S - 4
 (c) P - 2, Q - 1, R - 3, S - 4
 (d) P - 2, Q - 1, R - 4, S - 3

Match the following :

List-I

List-II

(P) 2.220

1. $2 + \frac{2}{100}$

(Q) 2.022

2. $2 + \frac{2}{10} + \frac{2}{100}$

(R) 2.020

3. $20 + \frac{2}{10} + \frac{2}{1000}$

(S) 20.202

4. $2 + \frac{2}{100} + \frac{2}{1000}$

- (a) P - 4, Q - 2, R - 3, S - 1
 (b) P - 4, Q - 2, R - 1, S - 3
 (c) P - 2, Q - 4, R - 3, S - 1
 (d) P - 2, Q - 4, R - 1, S - 3

4. Match the following :

List-I

List-II

(P) The snowfall during three winter months was 30.5 cm, 45.25 cm, and 25.25 cm. Then the total snowfall (in cm) in these three months is

1. 7.095

(Q) Kritika's weight is 35 kg 75 g and her sister Khushi's weight is 27 kg 980 g. Kritika is heavier than Khushi by (in kg)

2. 12.75

(R) Naresh drives 42 km 35 m in the morning and 25 km 7 m in the evening. The distance he drives in all (in km) is

3. 101

(S) Anu had ₹ 218.50. She bought some ice-creams for ₹ 205.75. The money she have now (in ₹) is

4. 67.042

- (a) P - 1, Q - 3, R - 2, S - 4
 (b) P - 3, Q - 1, R - 2, S - 4
 (c) P - 3, Q - 1, R - 4, S - 2
 (d) P - 1, Q - 3, R - 4, S - 2

Assertion & Reason Type

Directions: In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statement, mark the correct answer as

- (a) If both assertion and reason are true and reason is the correct explanation of assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of assertion.
 (c) If assertion is true but reason is false.
 (d) If assertion is false but reason is true.

1. **Assertion** : While subtracting $2\frac{5}{10}$ from $4\frac{1}{10}$, borrowing is needed.

Reason : $\frac{1}{10}$ is less than $\frac{5}{10}$ so, we borrow 1 whole = $\frac{10}{10}$ from the integer part.

2. **Assertion**: The fraction $\frac{145}{100}$ can be written as $1\frac{4}{10} \frac{5}{100}$.

Reason : 145 hundredths = 1 unit, 4 tenths, and 5 hundredths.

3. **Assertion** : 1.005 is closer to 1 than 1.05.
Reason : The thousandths place digit in 1.005 is smaller than the hundredths place digit in 1.05.

4. **Assertion** : 3.5 hours = 3 hours 50 minutes.
Reason : In decimal, 0.5 hour means 30 minutes.

5. **Assertion** : The sequence 1.5, 1.4, 1.3, 1.2 is increasing.
Reason : In each step, 0.1 is being subtracted.

6. **Assertion** : On subtracting 5 km 8 m from 26 km 50 m, the result will be obtained as 21.042 km.
Reason : 26 km 50 m and 5 km 8 m can be expressed in km as 26.500 km and 5.080 km respectively.

7. **Assertion** : $714.371 - 54.21$ is same as $714.371 - 54.021$.
Reason : To subtract two decimals, they should be first converted into like decimals.

8. **Assertion** : The number 211.902 is expressed in expanded form as $2 \times 100 + 1 \times 10 + 1 + 9 \times \frac{1}{10} + 2 \times \frac{2}{100}$.

Reason : As we go from left to right, the multiplying factor becomes $\frac{1}{10}$ of the previous factor.

9. **Assertion** : The sum of 27.8, 175.09 and 685.7 is 888.59.

Reason : To subtract decimal numbers, we place them columnwise correctly in the place value table and subtract them in the same way as of whole numbers.

10. **Assertion** : Expressing 735 m into km, we get 735000 km.

Reason : $1 \text{ m} = \frac{1}{1000} \text{ km}$



Comprehension Type

PASSAGE-I : Riya, Meher, and Aarav are participating in a school Bake-Off. They are baking layered cakes and measuring their ingredients precisely using mixed numbers (whole numbers with tenths and hundredths of a unit). Each of them has a different cake plan.

- Riya plans to use $3\frac{9}{10}$ kg of flour.

- Meher needs $2\frac{7}{10}$ kg of flour.

- Aarav brings $1\frac{55}{100}$ kg of flour from home and borrows $1\frac{3}{10}$ kg from Riya.

After baking :

- Riya uses $2\frac{6}{10}$ kg of flour.
- Meher uses $2\frac{3}{10}$ kg of flour.
- Aarav uses all the flour he has.

- Who used the highest quantity of flour?
 (a) Riya (b) Meher
 (c) Aarav (d) All used the same.
- How much flour is Riya left with after using some and lending to Aarav?
 (a) $\frac{1}{10}$ kg (b) $\frac{1}{2}$ kg
 (c) $\frac{5}{10}$ kg (d) None of these
- How much total flour did Meher left after baking?
 (a) $\frac{9}{10}$ kg (b) $3\frac{85}{100}$ kg
 (c) $3\frac{45}{100}$ kg (d) $\frac{4}{10}$ kg

PASSAGE-II : To change a decimal numeral to a fraction, express the decimal as a fraction with denominator 10, 100, 1000 and then reduce it to its lowest term.

- The sum of 67.25, 5.5, 9.85 in fraction form is
 (a) $\frac{826}{100}$ (b) $\frac{826}{10}$ (c) 207 (d) $\frac{820}{100}$
- Amit bought a comic book for ₹ 45.60 and a magazine for ₹ 70.00. What is the total amount (in ₹) spent by Amit (express in fractional form)?
 (a) $\frac{102}{5}$ (b) $\frac{156}{2}$ (c) $\frac{1156}{10}$ (d) $\frac{1156}{5}$

3. Convert 19.07 kg into grams.

- (a) 19070 g (b) $\frac{1907}{10}$ g
(c) 1907 g (d) None of these

PASSAGE-III : A squash ball weighs 34000 mg, a cricket ball weighs 45 g, a basketball weighs 0.65 kg and a football weighs 3 kg.

- Which of the given sports balls is lightest in weight?
(a) Cricket ball (b) Football
(c) Squash ball (d) Basketball
- What is the total weight (in kg) of basketball and football?
(a) 0.365 kg (b) 3.65 kg
(c) 365 kg (d) 3 kg
- What is the total weight of all the balls (in mg)?
(a) 3729 mg (b) 3.729 mg
(c) 3729000 mg (d) 0.3729 mg

Subjective Problems

Very Short Answer Type

- Show the number $15\frac{3}{100}$ on the number line.
- What mistake happens if you read 0.06 kg as 0.6 kg?
- How many hundredths are there in 1 unit?
- In the sequence 3.4, 3.7, 4.0, __, what is the next term?
- Which is larger $5\frac{3}{100}$ or $\frac{503}{100}$?
- What is the value of $17.564 + 12.387 + 23.091$?
- 0.123 or 0.125, which is greater?
- What is the sum of 12.17, 7.83 and 16.2?
- Express 46 rupees 87 paise as rupees using decimals.
- Subtract 0.25 from 0.37.
- Express $6 \times \frac{1}{10} + 8 \times \frac{1}{100}$ as a decimal.

12. Simplify : $6 - 0.4 - 0.87$.

13. What is the difference between 1.16 and the sum of 6.12 and 2.43?
14. Show 1.8 on the number line.
15. Write in order from least to the greatest 2.1, 2.105, 2.011.

Short Answer Type

- Write the number names of the following numbers in tenth and hundredth.
(a) $12\frac{3}{10}\frac{23}{100}$ (b) $\frac{1}{10}\frac{5}{100}$
- Write the given numbers in ascending order.
 $1\frac{2}{10}\frac{3}{100}$, $\frac{13}{10}\frac{23}{100}$, $\frac{113}{100}$
- The given sequence is: 2.6, 2.9, 3.2, 3.5. Identify the pattern and write the next two terms in the sequence.
- Find the sum of 40 rupees 80 paise and 95 rupees 50 paise.
- Ramesh and Rahul walked back to their homes by walking 3.2 km and 3.58 km respectively. Who walked more distance and by how much?
- Simplify: $3.23 + 29.375 + 81.2016 + 0.3376$
- A man purchases an almirah for ₹ 3675. He pays ₹ 172.50 as its cartage and spends ₹ 64.80 on its repair. How much does he spend for the almirah?
- The height of Jyoti and Geetanjali is 1.59 m and 0.94 m, respectively. How much shorter is Geetanjali than Jyoti?
- Simplify: $53.5 - 34.68 + 64.75 - 28.9$.
- From a ribbon of length 15 m, two pieces of 7 m and 6.5 m were cut. Find the length of remaining piece.

Long Answer Type

- Express the following decimal numbers as sum of tenths, hundredths, and thousandths

- (a) 1.24 (b) 2.365
(c) 0.432 (d) 0.04

2. Aman is preparing for a cycling competition. Every day, he increases the distance he cycles by a fixed decimal amount. The distance he cycled over the first five days is recorded as follows:

Day 1: 3.5 km Day 2: 4.1 km
Day 3: 4.7 km Day 4: 5.3 km
Day 5: 5.9 km

- (a) Identify the pattern in the sequence. What decimal number is being added each day?
(b) Write the distances for Day 6, Day 7, and Day 8.
(c) How much total distance does Aman cycle over the 8 days?
3. Simplify: (i) $(503.37 + 253) - (81.61 - 41.4135)$
(ii) $91.469 + 39.65 - 57.255 + 14.50$
(iii) $23.667 - 2.55 - 19.205 + 80.432$
4. Rajendra purchased 6 kg 25 g of fruits and 4 kg 355 g of vegetables and put them in a bag. If this bag with these contents weighs 11 kg, find the weight of the empty bag.
5. The distance between Raj's house and his office is 16 km. He covers 11 km 35 m by car, 3 km 75 m by bus and the rest on foot. How much distance does he cover by walking?

Integer/Numerical Value Type

1. In the pattern: 2.5, 2.0, 1.5, 1.0, 0.5, _____. How many steps of subtraction of 0.5 are needed to reach zero?
2. A rod is 3.5 feet long. Its length in inches is _____ inches.
3. The sum of $2\frac{4}{10}\frac{64}{100}$ and $3\frac{6}{10}\frac{36}{10}$ is _____.
4. A wooden plank is $7\frac{6}{10}$ m long. A piece of $\frac{60}{100}$ m is cut off, then the remaining length in meters is _____.

5. The digit at tenths place in the sum, if we add 8.3 and 5.6 is _____.
6. The sum of the digits in the product of the sum and the difference of 4.5 and 2.5 is _____.
7. The whole number part in the sum of $3 + 0.4 + 0.01 + 0.001$ is _____.
8. The height of Sonu is 1.25 m and that of Reena is 1.3 m. Reena is _____ cm taller than Sonu.
9. The tenths place digit in the sum, if we add 13.5 and 1.57 is _____.
10. The value of P , if $0.04 = \frac{P}{25}$ is _____.



Case Based Questions

Case I : In Ms. Ananya's science lab, students were performing an experiment measuring different volumes of a chemical solution using precise decimal readings. However, a mix-up occurred when Rohan noted some measurements using fractional units, while Priya used decimals, and Arjun used dot notation (which looked like decimal points but actually separated mL and drops).

Here are the records of their readings:

Rohan: Recorded volumes as: $2\frac{3}{10}$ mL, $2\frac{4}{10}$ mL, $2\frac{5}{10}$ mL

Priya: Wrote down 2.3 mL, 2.4 mL, 2.5 mL

Arjun: Recorded the third reading as 2.50 mL thinking it meant 2 mL and 50 drops.

Their teacher explained that:

- 10 drops = 1 mL
- A difference between 2.3 mL and 2.30 mL can confuse if not understood well.
- A proper pattern is seen in the decimal increments, which help to forecast future values in measurements.

Based on the above information, answer the following questions.

1. What common mistake did Arjun make in writing "2.50 mL"?

- (a) He converted drops to mL correctly
- (b) He used dot notation incorrectly, confusing it with decimal notation
- (c) He added too many zeroes
- (d) He converted mL to L incorrectly.

2. Which of the following best describes the pattern in Rohan's readings:

$$2\frac{3}{10} \text{ mL}, 2\frac{4}{10} \text{ mL}, 2\frac{5}{10} \text{ mL}$$

- (a) Increases by 0.2 mL
- (b) Increases by $\frac{1}{10}$ mL
- (c) Increases by $\frac{2}{10}$ mL
- (d) Decreases by 0.1 mL

3. What is the correct decimal equivalent of Rohan's $2\frac{5}{10}$ mL?

- (a) 2.05
- (b) 2.5
- (c) 5.2
- (d) 2.15

4. Why can "2.30" and "2.3" be considered the same in this experiment?

- (a) Because both represent 2.3 millilitres in volume
- (b) Because 2.30 is greater than 2.3
- (c) Because extra zero changes the quantity
- (d) Because dot was added accidentally

What will be the next two terms if the decimal volume continues as: 2.3 mL, 2.4 mL, 2.5 mL...

- (a) 2.6 mL, 2.7 mL
- (b) 2.8 mL, 2.9 mL
- (c) 2.55 mL, 2.60 mL
- (d) 2.4 mL, 2.3 mL

Case II : During Covid-19 Lockdown, Reena's husband lost his job. It becomes very difficult for the family to survive, due to lack of essential things. So, Reena decided to sale her gold jewellery that she had. When goldsmith did the weight of a necklace, a pair of earrings and a nosepin, he found the weight was 71.52 g, 6.39 g and 0.154 g respectively.



Based on the above information, answer the following questions.

1. Find the sum of the weight of gold jewellery.

- (a) 77.064 g
- (b) 0780.64 g
- (c) $\frac{9758}{25}$ g
- (d) $\frac{9758}{125}$ g

2. The smallest and largest weights of gold jewellery items expressed as fraction, are respectively.

- (a) $\frac{154}{100}$ g, $\frac{71520}{100}$ g
- (b) $\frac{639}{100}$ g, $\frac{154}{1000}$ g
- (c) $\frac{154}{1000}$ g, $\frac{7152}{100}$ g
- (d) $\frac{154}{100}$ g, $\frac{7152}{100}$ g

3. Find the difference between the weight of necklace and nosepin in fraction form.

- (a) $\frac{71520}{1000}$ g
- (b) $\frac{35683}{500}$ g
- (c) $\frac{71520}{100}$ g
- (d) $\frac{35683}{1000}$ g

4. How much weight of necklace is more than the weight of nosepin (in decimal form)?

- (a) 71.366 g
- (b) 78.054 g
- (c) 35.683 g
- (d) 71.307 g

5. If the cost of 1g of gold is ₹ 5000, then amount got by Reena from goldsmith is

- (a) ₹ 3,20,000
- (b) ₹ 4,00,020
- (c) ₹ 3,50,000
- (d) ₹ 3,90,320

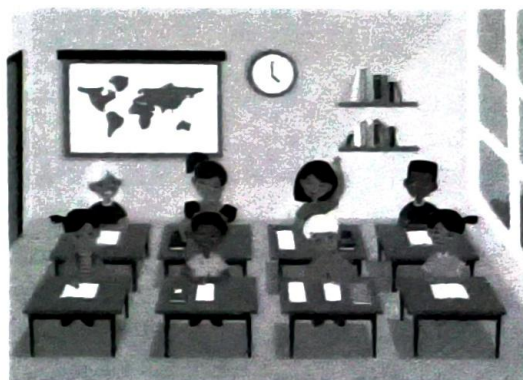
Case III : Rajat's mother send him to purchase 1500 g of corn flour and 250 g of sugar to prepare some sweets. When he went to shopkeeper, the shopkeeper gave him two packets weighing 0.970 kg and $\frac{1}{2}$ kg of corn flour and a packet weighing $\frac{22}{125}$ kg of sugar.



Based on the above information, answer the following questions.

- Weight of corn flour that Rajat received (in kg), is
(a) 1.560 kg (b) 2 kg
(c) 1.320 kg (d) 1.470 kg
- Weight of sugar that Rajat received (in kg), is
(a) 0.157 kg (b) 0.176 kg
(c) 2.76 kg (d) 2 kg
- Sum of weight of things that Rajat asked to shopkeeper, is
(a) 1.750 kg (b) 2.370 kg
(c) 2.570 kg (d) 1.370 kg
- Sum of weight of things that Rajat received from shopkeeper, is
(a) 1.350 kg (b) 1.750 kg
(c) 2.70 kg (d) 1.646 kg
- How much less quantity of things that Rajat received from shopkeeper?
(a) 0.300 kg (b) 3.750 kg
(c) 0.104 kg (d) 1.750 kg
- Total length of cloth required by tailor to make complete dress is
(a) 5.56 m (b) 5.36 m
(c) 4.36 m (d) 4.56 m
- Length of cloth that tailor had (in m), is
(a) 3.56 m (b) 1.46 m
(c) 4.42 m (d) 1.14 m
- How much more cloth is needed by tailor to stitch complete dress?
(a) 1.56 m (b) 4.32 m
(c) 2.16 m (d) 1.14 m

Case V : In the previous class, teacher taught the students about "Decimals" and students were very excited to learn it and took part in every activity arranged by teacher to teach this concept.



Now, the teacher wants to check that, students really grasped the concept or not. Thus, she asked them certain questions.

- Which of the following stands third on arranging the following decimals in descending order?
23.6, 46.32, 20.09, 22.267, 32.306
(a) 22.267 (b) 23.6
(c) 32.306 (d) 20.09
- Which of the following is the smallest decimal?
(a) 70.6 (b) 80.8
(c) 7.06 (d) 6.80
- Which of the following stands second last on arranging the following decimals in ascending order?
8.60, 6.9, 0.690, 0.906, 0.96, 8.06, 0.096
(a) 6.9 (b) 0.690
(c) 8.06 (d) 0.096

Case IV : Riya and her mother wants to attend a marriage. Therefore, Riya's mother went to tailor to prepare a designer dress for her daughter. Tailor told that he needed 2 m 52 cm of cloth to stitch upper part of dress and 3 m 4 cm of cloth to stitch bottom part of dress. But the tailor had only 4 m 42 cm of cloth with him.



Based on the above information, answer the following questions.

- Cloth required to stitch the upper part of dress (in m), is
(a) 1.56 m (b) 2.36 m
(c) 2.52 m (d) 1.63 m
- Cloth required to stitch the bottom part of the dress (in m), is
(a) 2.04 m (b) 1.04 m
(c) 4.03 m (d) 3.04 m

4. Which of the following is the greatest number?

(a) $1 + \frac{2}{100} + \frac{4}{1000}$

(b) $10 + \frac{2}{10} + \frac{4}{1000}$

(c) $10 + 2 + \frac{4}{1000}$

(d) $\frac{1}{10} + \frac{2}{100} + \frac{4}{1000}$

5. Which of the following is the smallest number?

(a) $\frac{5}{10} + \frac{9}{1000}$

(b) $\frac{5}{100} + \frac{9}{1000}$

(c) $5 + \frac{9}{10}$

(d) $\frac{9}{100} + \frac{5}{1000}$
